

```

#include<iostream>
using namespace std;
int main()
{
    int n;
    cout<<"Enter n:";
    cin>>n;
    int a[n]={0};//Initialize all elements of the array to zero.
    //input n elements
    for(int index=0;index<n;index++)
    {
        cout<<"\n Enter at a["<<index<<"]:";
        cin>>a[index];
    }
    //print element by element using index
    for(int index=0;index<n;index++)
    {
        cout<<"\n at a["<<index<<"]:"<<a[index];//Will print garbage values because
initially there will be garbage.
    }
    return 0;
}

```

```

for(int item:a)//The type of array and the type of variable have to match.
{
    cout<<"\n"<<item;
}

```

```

#include<iostream>
//An array is given. Reverse the array without using another array.
using namespace std;
int main()
{
    int n=5;
    int a[n]={11,22,33,44,55};//Initialize all elements of the array to zero.
    cout<<"\nArray consists of :\n";

    for(int index=n-1;index>=0;index--)
    {
        cout<<"\n at a["<<index<<": "<<a[index];//Will print garbage values because
initially there will be garbage.
    }
    //reverse

    cout<<"\nAfter Reversed Array consists of :\n";
    for(int index=n-1;index>=0;index--)
    {
        cout<<"\n at a["<<index<<": "<<a[index];//Will print garbage values because
initially there will be garbage.
    }

    return 0;
}

```

```

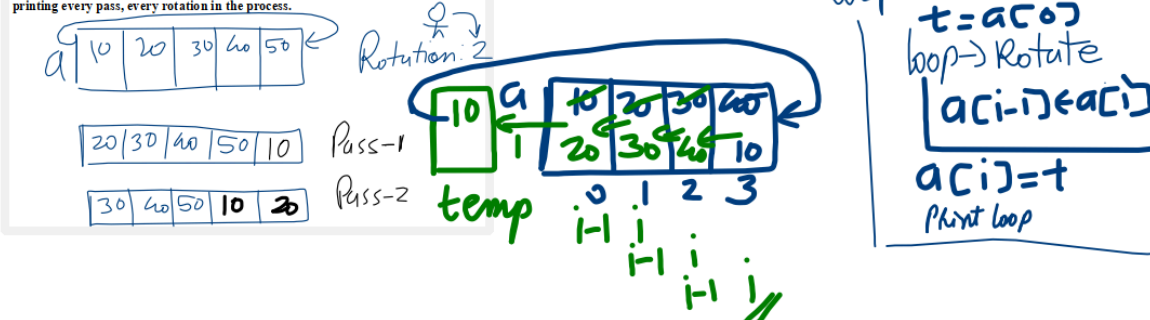
#include<iostream>
//Find minimum and maximum in the given array, along with the position of each.
using namespace std;
int main()
{
    int a[]={66,11,55,22,99,88,77,12,45,67,89,34,28,19};
    int size=sizeof(a)/sizeof(a[0]);
    //Size of entire array / size of one block
    //56/4-->14
    cout<<"\nArray length is:"<<size;
    //create references for min,min_position,max,max_position
    min=max=a[0];
    min_position=max_position=0;
    for(int index=1;index<size;index++)
    {
        //If a[index] is less than `minimum`, update `minimum` and `minimum_position`.

        //If a[index] is more than `maximum`, update `maximum` and `maximum_position`.
    }
    //Print minimum and position of minimum.
    //Print the maximum and the position of the maximum.

    return 0;
}

```

A user has been provided with an array of five elements. Accept from the user a number of types 'rotation' and perform a clockwise rotation in the given array, printing every pass, every rotation in the process.



```
#include<iostream>
```

```
//A user has been provided with an array of five elements. Accept from the user a number of
types `rotation` and perform a clockwise rotation in the given array,
```

```
// printing every pass, every rotation in the process.
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int a[]={11,22,33,44,55};
```

```
    int size=sizeof(a)/sizeof(a[0]);
```

```
    cout<<"\nArray length is:"<<size;
```

```
    int rotations;
```

```
    cout<<"\nEnter the number of times rotation is needed: ";
```

```
    cin>>rotations;
```

```
    for(int r=1;r<=rotations;r++)//For one rotation at a time
```

```
    {
```

```
        int temp=a[0];//Creating buffer
```

```
        for(int index=1;index<size;index++)//Physically performs their rotation part
```

```
        {
```

```
            a[index-1]=a[index];
```

```
        }
```

```
        a[size-1]=temp;
```

```
        //print
```

```
        cout<<"\nAfter Pass "<<r<<":";
```

```
        for(int item:a)
```

```
        {
```

```
            cout<<item<<" ";
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```

Understanding 2D matrix

```
#include<iostream>
```

```
//A user has been provided with an array of five elements. Accept from the user a number of  
types `rotation` and perform a clockwise rotation in the given array,
```

```
// printing every pass, every rotation in the process.
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int rows,columns;
```

```
    cout<<"Enter dimensions of 2D array\n";
```

```
    cin>>rows>>columns;
```

```
    int m[rows][columns];
```

```
    //reading
```

```
    for(int r=0;r<rows;r++)
```

```
    {
```

```
        for(int c=0;c<columns;c++)
```

```
        {
```

```
            cout<<"Enter data for m ["<<r<<"]["<<c<<"]:";
```

```
            cin>>m[r][c];
```

```
        }
```

```
    }
```

```
    //printing
```

```
    cout<<"\n Matrix has:\n";
```

```
    for(int r=0;r<rows;r++)
```

```
    {
```

```
        for(int c=0;c<columns;c++)
```

```
        {
```

```
            cout<<m[r][c]<<"\t";
```

```
        }
```

```
        cout<<endl;//In order to move to the next line
```

```
    }
```

```
    return 0;
```

```
}
```

```

#include<iostream>
//Print transpose of a matrix.
using namespace std;
int main()
{
    int rows,columns;
    cout<<"Enter dimensions of 2D array\n";
    cin>>rows>>columns;
    int m[rows][columns];
    //reading
    for(int r=0;r<rows;r++)
    {
        for(int c=0;c<columns;c++)
        {
            cout<<"Enter data for m ["<<r<<"]["<<c<<"]:";
            cin>>m[r][c];
        }
    }
    //printing
    cout<<"\n Matrix has:\n";
    for(int r=0;r<rows;r++)
    {
        for(int c=0;c<columns;c++)
        {
            cout<<m[r][c]<<"\t";
        }
        cout<<endl;//In order to move to the next line
    }

    //printing
    cout<<"\n Matrix Transpose has:\n";
    for(int c=0;c<columns;c++)
    {
        for(int r=0;r<rows;r++)
        {
            cout<<m[r][c]<<"\t";
        }
        cout<<endl;//In order to move to the next line
    }

    return 0;
}

```

```
#include<iostream>
//We are providing a matrix of 2, 3 with all the data given.
//Your job is to print the matrix along with the total of each row at the end of that row.
using namespace std;
int main()
{
    int rows=2,columns=3;
    int m[2][3]={{10,20,30},{1,2,3}};
    //In a static array one must declare the number of columns.
    //The row will be picked up automatically.
    //printing
    cout<<"\n Matrix has:\n";
    for(int r=0;r<rows;r++)
    {
        for(int c=0;c<columns;c++)
        {
            cout<<m[r][c]<<"\t";
        }
        cout<<endl;//In order to move to the next line
    }

    return 0;
}
```

```

#include<iostream>
using namespace std;
int main()
{
    string name;
    //cout<<"who are you :";
    //cin>>name;
    //cout<<"you entered:"<<name;
    //new for line of data

    cout<<"who are you :";
    getline(cin,name);
    cout<<"you entered:"<<name;
    return 0;
}

```

```

#include<iostream>
using namespace std;
int main()
{
    string name,from;
    cout<<"where are you from:";
    cin>>from;
    cin.ignore();
    cout<<"\nyou are from :"<<from;
    cout<<"\nwho are you :";
    getline(cin,name);
    cout<<"you entered:"<<name;
    return 0;
}

```

```

#include<iostream>
#include<string>
using namespace std;
int main()
{
    string line;
    cout<<"Enter a line:";
    getline(cin,line);
    cout<<"\nyou entered:"<<line;
    cout<<"\nsizeof:"<<sizeof(line);
    cout<<"\nsize:"<<line.size()<<"Returns the total number of characters. ";
    cout<<"\nlength:"<<line.length();
}

```

```

        for(int i=0;i<line.length();i++)
        {
            cout<<"\ni="<<i<<": "<<line[i];
        }

        return 0;
    }
//Count vowels.
#include<iostream>
#include<string>
//vowels:a e i o u
using namespace std;
int main()
{
    string line;
    int count=0;
    cout<<"Enter a line:";
    getline(cin,line);
    //count vowel
    for(char c:line)
    {
        if(c=='a' || c=='e' || c=='i' || c=='o' || c=='u')
            count++;
    }
    cout<<"\nTotal vowels in line are:"<<count;
    return 0;
}

```

//Use the above program logic to count the frequency of a character given by the user.

```

#include<iostream>
#include<string>
//use of +
using namespace std;
int main()
{
    /*
    string word1,word2,word3;
    word1="am";
    word2="ar";
    word3=word1+word2;//The `+` symbol is used to concat one string to another.
    cout<<word3;*/
    string s1="",s2="abcdefgh";
    for(char c:s2)
    {
        s1+=c;
        cout<<s1<<"\n";
    }
}

```

```

    }
    return 0;
}

```

```

#include<iostream>
#include<string>
//We want a given line of string in such a manner that the first character of every word
becomes capital.
//input:hl THIS is IT
//output:Hi This Is It
using namespace std;
int main()
{
    cout<<(int)'a';
    cout<<endl<<(char)toupper('a');//Dynamic casting.
    cout<<endl<<(char)tolower('A');//Dynamic casting.
    //The process in which an `int` can be cast to `char` or `char` can be cast to an
integer

```

```

/*
    string word,rword="";
    cout<<"Enter word:";
    cin>>word;
    for(int i=word.length()-1;i>=0;i--)
    {
        rword+=word[i];
    }
    if(word==rword)
        cout<<endl<<word<<" Is palindrome ";
    else
        cout<<endl<<word<<" Is not palindrome ";*/
    return 0;
}

```

```

//Return first non-repeating alphabet.
//word:assassination. .
#include <iostream>
#include <string>
//Take a line of string. Print the frequency of every alphabet.
using namespace std;

int main()
{
    string s;

    cout << "Enter a string: ";
    getline(cin, s);

    int frequency[256]={0}; //256 stands for all 256 characters ASCII can take.

    for(char c:s)
    {
        frequency[(unsigned int)c]++;
    }

    /*for(int i:frequency)
    {
        cout<<endl<<i<<" ";
    }*/

    for(char c:s)
    {
        cout<<endl<<c<<":"<<frequency[(unsigned int)c];

    }

    return 0;
}

```

```
#include <iostream>
#include <string>
//Take a line of string. Print the frequency of every alphabet.
using namespace std;
```

```
int main()
{
    int a[]={7,1,2,9,3,6,4,8,5};
    cout<<"Array:";
    for(int i:a)
        cout<<i<<" ";
    for(int i=0;i<sizeof(a)/sizeof(a[0]);i++)
    {
        for(int j=0;j<sizeof(a)/sizeof(a[0])-1;j++)
        {
            if(a[j]>a[j+1])
            {
                int temp=a[j];
                a[j]=a[j+1];
                a[j+1]=temp;
            }
        }
        cout<<"\nSorted Array:";
    }
    for(int i:a)
        cout<<i<<" ";
```

```
    return 0;
```

```
}
```

//Accept 10 names from the user and sort them in ascending order using the above algorithm.

```
#include <iostream>
#include <string>
//Take a line of string. Print the frequency of every alphabet.
using namespace std;
```

```
int main()
{
    string a[]{"zian", "wonder", "amar", "ajay", "kunal" };
    cout<<"Array:";
    for(string i:a)
        cout<<i<<" ";
    for(int i=0;i<sizeof(a)/sizeof(a[0]);i++)
    {
        for(int j=0;j<sizeof(a)/sizeof(a[0])-1;j++)
        {
            if(a[j]>a[j+1])
```

```
        {
            string temp=a[j];
            a[j]=a[j+1];
            a[j+1]=temp;
        }
    }
    cout<<"\nSorted Array:";
    for(string i:a)
        cout<<i<<" ";

    return 0;
}
```